ɔn 1: To be answered in **(10) Minutes** [ ] / 2 Marks

Will the **Max-Heap** look like, both as an **array** and a **binary tree**, after running the **Build Max-Heap** thm on the following array: **[5, 3, 8, 4, 1, 7, 9]**?

Array [1 mark]	Binary tree [1 mark]

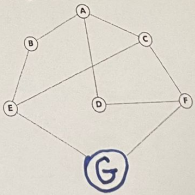
Question 2: To be answered in **(20) Minutes** [ ] / 3 Marks

Choose one correct answer for each statement and fill the following table:

Question	1	2	3	4-1	4-2	5
Answer						

1. ***Heap Sort is:***
   a. *Stable and in-place*
   b. *Unstable and in-place*
   c. *Stable and not in-place*
   d. *Unstable and not in-place*

2. **Given the graph below, starting from vertex (A) give the BFS traversal assuming alphabetical order traversal:**
   a. A → B → C → D → E → F → G
   b. A → C → D → B → F → E → G
   c. A → B → D → C → G → E → F
   d. A → D → C → B → F → E → G



Name (in English): _____ Student ID: _____

3. **Find the MST and its weight algorithm for the graph G:**
   a. MST = {A → C, B → C}, MST = 6
   b. MST = {A → B, A → C, C → D}, MST = 8
   c. MST = {A → B, B → C, C → D}, MST = 6
   d. MST = {A → B, C → D}, MST = 4

Graph G:

From	To	Weight
A	B	1
A	C	4
B	C	2
B	D	6
C	D	3

4. **For the same graph G in (3) answer the following:**
   4.1. What is the order of nodes visited using Dijkstra's algorithm starting from vertex A?
      a. A → B → C → D
      b. A → C → B → D
      c. A → B → D → C
      d. A → C → D → B

   4.2. What is the shortest path from A to D using Dijkstra's algorithm?
      a. A → B → D (Cost: 7)
      b. A → C → D (Cost: 7)
      c. A → B → C → D (Cost: 6)
      d. A → D (Cost: 10)

5. **Which of the following is NOT a property of the Bellman-Ford algorithm?**
   a. It works with graphs containing negative weights.
   b. It detects negative weight cycles.
   c. It can find the shortest path from a single source to all vertices.
   d. It uses a priority queue for optimization.

End of Questions
Good Luck